COER UNIVERSITY

DBMS LAB ASSIGNMENT | LAB SHEET 1: DATA DEFINITION LANGUAGE (DDL)

Task 1: Database & Table Creation

Question 1.a Question: Create a database named Company.

Answer:

Explanation: The CREATE DATABASE statement creates a database logical container named Company in MySQL. The USE statement selects Company as the active database for subsequent operations.

Question 1.b Question: Create a table named Employees with the following structure: EmpID (INT, AUTO_INCREMENT, PRIMARY KEY), FirstName (VARCHAR(50), NOT NULL), LastName (VARCHAR(50)), Department (VARCHAR(50)), Salary (DECIMAL(10,2)), Email (VARCHAR(100), UNIQUE).

Answer:

Explanation: This statement defines the Employees schema. EmpID automatically increments for new rows and serves as the primary key. FirstName is mandatory (NOT NULL), and Email enforces uniqueness across records.

Task 2: DDL Queries on the Employees Table

Question 2.a Question: Describe the structure of the table.

Answer:

Explanation: The DESCRIBE command displays column names, data types, nullability, keys, default values, and auto-increment properties of the table.

Question 2.b Question: Rename the table to Staff.

Answer:

Explanation: RENAME TABLE changes the table name from Employees to Staff in the database catalog without altering stored data or structure.

Question 2.c Question: Add a new column PhoneNumber (VARCHAR(15)).

Answer:

Explanation: ALTER TABLE ... ADD appends a new column named PhoneNumber to the existing table definition.

Question 2.d Question: Modify the data type of Salary to DECIMAL(12,2).

Answer:

Explanation: ALTER TABLE ... MODIFY changes the column precision of Salary to support numbers up to 12 total digits with 2 decimal places.

Question 2.e Question: Add a DEFAULT value of 'General' to the Department column.

Answer:

Explanation: Setting a DEFAULT constraint ensures that if a record is inserted without a department, MySQL automatically assigns 'General'.

Question 2.f Question: Drop the PhoneNumber column.

Answer:

Explanation: ALTER TABLE ... DROP COLUMN permanently removes the PhoneNumber attribute from the table.

Question 2.g Question: Add a CHECK constraint to ensure Salary is greater than 0.

Answer:

Explanation: Adds a domain constraint named chk_Salary to validate that any inserted or modified salary value is strictly greater than 0.

Question 2.h Question: Rename the column FirstName to GivenName.

Answer:

Explanation: RENAME COLUMN changes the column identifier from FirstName to GivenName while maintaining its data type and constraints.

Question 2.i Question: Drop the chk_Salary constraint.

Answer:

Explanation: Removes the chk_Salary check constraint from the table definition.

Question 2.j Question: Create an index on the Email column.

Answer:

Explanation: Creates a B-tree index named idx_email on the Email column to speed up search and query operations.

Question 2.k Question: Drop the index on the Email column.

Answer:

Explanation: DROP INDEX deletes the specified index structure from the table.

Question 2.l Question: Truncate the table to remove all rows without affecting the structure.

Answer:

Explanation: TRUNCATE TABLE removes all records from the table quickly and resets auto-increment counters, while keeping the table structure intact.

Question 2.m Question: Drop the Staff table completely.

Answer:

Explanation: DROP TABLE deletes the table definition, data, constraints, and indexes permanently from the database.

Task 3: Conceptual Questions

Question 3.1 Question: What is the difference between TRUNCATE and DELETE?

Answer:

1. Command Category: TRUNCATE is a DDL command, whereas DELETE is a DML command. 2. Speed & Efficiency: TRUNCATE is faster as it deallocates memory pages directly; DELETE removes rows one by one. 3. WHERE Clause: DELETE supports conditions using WHERE; TRUNCATE removes all rows unconditionally. 4. Auto-increment Reset: TRUNCATE resets AUTO_INCREMENT counters back to 1; DELETE retains the current counter value.

Explanation: Understanding the distinction helps in choosing the right command based on performance needs and transaction safety.

Question 3.2 Question: Explain the use of the CHECK constraint.

Answer:

The CHECK constraint enforces domain integrity by ensuring that all values entered in a column satisfy a logical boolean condition (e.g., Salary > 0 or Age >= 18). If an insert or update violates the condition, MySQL throws an error and rejects the operation.

Explanation: CHECK constraints prevent invalid data entry at the database level.

Question 3.3 Question: Why is it important to define a PRIMARY KEY for a table?

Answer:

1. Uniqueness: Guarantees that every row in the table can be uniquely identified. 2. Entity Integrity: Ensures key columns cannot contain NULL values. 3. Performance: MySQL creates a clustered index on the primary key for fast row lookup. 4. Relationships: Acts as a reference key for foreign keys in other tables.

Explanation: Primary keys maintain fundamental database integrity and optimize search queries.

Question 3.4 Question: How does ALTER TABLE help in modifying an existing table?

Answer:

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ALTER TABLE enables structural changes to an existing table without deleting stored data. It supports adding
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Explanation: It allows database schemas to evolve over time without losing existing records.

Question 3.5 Question: What is the purpose of creating an index in a table?

Answer:

An index builds a search pointer structure (B-tree) on specified columns to significantly accelerate record retrieval during SELECT queries (WHERE, JOIN, ORDER BY), eliminating the need for full table scans.

Explanation: Indexes trade small storage space and insert overhead for much faster reading performance.

Task 4: Exercises

Question 4.1 Question: Create a table named Projects (ProjectID INT PRIMARY KEY AUTO_INCREMENT, ProjectName VARCHAR(100) NOT NULL, StartDate DATE, EndDate DATE, Budget DECIMAL(15,2) CHECK (Budget > 0)). Perform at least 5 DDL operations on Projects, and write a query to retrieve column names.

Answer:

Explanation: Creates the Projects table with constraints, executes 5 DDL operations (ADD, MODIFY, RENAME COLUMN, CREATE INDEX, DROP INDEX), and queries system metadata to display table columns.